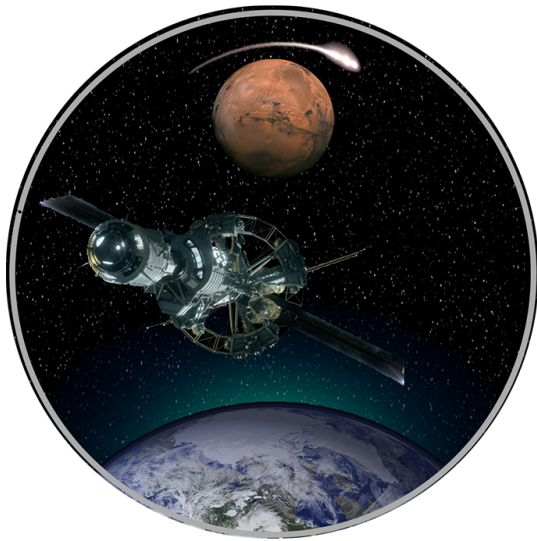




# EDEN

THE MANNED MISSION TO MARS

## Executive Summary

Manned Mars Mission Design GDP

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## Scientific Objectives

The purpose of a manned mission to Mars is two-pronged. The first objective would be conduct planetary science studies such as the search for life on Mars and the study of the Martian climate conditions and evolution of the Martian geological features. The second objective would be prepare for sustained human presence, with the crew studying how the human body would respond to the harsh conditions on Mars, as well as the reliability of converting local resources to useful products.

## Mission Profile and Trajectory

From the above objectives, the mission profile was developed.

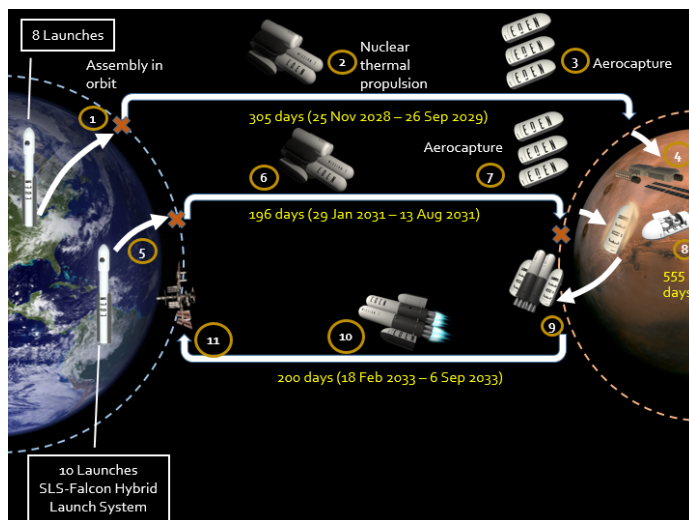
**T-2 Mission** The habitat cargo, In-Situ Resource Utilisation (ISRU) system and Mars Ascent Vehicle (MAV) will be launched up to Low Earth Orbit (LEO) in 8 separate launches, and assembled in LEO. This mission will leave Earth on 25 November 2028, two years prior to the crewed mission. This was to ensure a safe and functional living environment on Mars. The interplanetary transport for this mission will take 305 days and the capsules will separate prior to aerocapture and descent onto the Mars surface on 26 September 2029. Upon arrival on the surface of Mars, the ISRU system will begin making propellant for the MAV, and

upon confirmation of the successful production of the propellant, the T manned mission will be prepared.

**T Mission** Two orbiter capsules and an additional habitat capsule, along with the propulsion system, will be launched over 10 launches into LEO for assembly before leaving for Mars on 29 January 2031. The journey will take 196 days, significantly lesser than that of the T-2 mission as the manned crew of six is only able to stay in space for a maximum of 200 days for health and safety reasons. This manned mission is expected to reach Mars on 13 August 2031, when the orbiter will remain in orbit while the crew descend to Mar's surface on the habitat capsule. The crew will stay on the Martian surface for 555 days to achieve the above-mentioned objectives before ascending on the MAV to return to the orbiter for return.

Departure from Mars is expected to take place on 18 February 2033, and the journey back to Earth is expected to take 200 days, with arrival on 6 September 2033. On return, the crew will rendezvous with the ISS and the orbiter will be left for subsequent missions.

**Method** These trajectories were determined to be the most efficient transfers for their purposes, after porkchop plots for every possible interplanetary transfer were generated.



**Interplanetary trajectory** The obtained interplanetary trajectories were put into GMAT, a high fidelity mission design software developed by NASA. A deep space manoeuvre was determined to refine the trajectory and target an appropriate altitude for aerocapture at Mars and Earth arrival. The inclination (INC) at Mars arrival was also targeted to the landing site on Mars, while the INC and Right Argument of Ascending Node (RAAN) were targeted at Earth arrival to match the International Space Station (ISS) orbit.

## Parking orbit determination and Manoeuvres

The Mars parking orbit was designed to pass exactly over the landing site at altitude of 250 km with a period of 1 sol to facilitate the abort to orbit scenario.

The Earth parking orbit upon departure needed to match the INC of the transfer trajectory and was selected to be circular with 400 km altitude, similar to the ISS orbit. The Earth arrival parking orbit needed to match the ISS orbit for rendezvous.

A manoeuvre is performed after aerocapture to raise the periapsis above the atmosphere to the parking orbit periapsis. For the elliptical Mars orbit, a plane change manoeuvre on RAAN is necessary to match the Argument of Periapsis (AOP). Phasing manoeuvres are then required to rendezvous with the ISS in the Earth's case and to be above the landing site for Mars. The table below summarises the total  $\Delta V$  requirements (in km/s) for each mission.

|                          | Habitat $\Delta V$ | Crew Out-bound $\Delta V$ | Crew In-bound $\Delta V$ |
|--------------------------|--------------------|---------------------------|--------------------------|
| LEO assembly             | 1.080              | 1.264                     | 0                        |
| Interplanetary injection | 3.682              | 3.684                     | 2.541                    |
| Parking orbit insertion  | 0                  | 0.508                     | 1.550                    |

## Crew Transportation Vehicle

**Conceptual Design Process** The conceptual design of the overall mission and of each individual component was researched and developed. It was clear from the start that the main design drivers had to be its modularity and reusability. The former was achieved by ensuring the fitness for launch of each mission constituent and by devising a simple and stable docking mechanism; the latter was met by a versatile capsule design, which could be utilised for different purposes, from carrying payload such as the MAV or habitat, to hosting the crew during travel. Moreover, the "core capsules", i.e. the central component carrying the Nuclear Thermal Propulsion (NTP) and docking mechanisms, were equipped with their own heat shields for

aerocapture in order to make them reusable for the second mission inbound trip. Several iterations of CAD models were produced and this procedure became crucial for team integration and communication, as well as problem identification at each design stage.

**Finite Element Modelling** Finite element analysis was performed on the crew capsule and the MAV. Eigenvalue and steady-state dynamic analyses were performed for the launch phase to determine the natural frequencies and maximum accelerations respectively. For the EDL phase, the lower part of the capsule was assumed to be subjected to a maximum dynamic pressure of 12.7 kPa. Mesh convergence was then performed and a maximum displacement of 4.2 cm was found at both sides of the capsule. In future iterations, deflections are expected to decrease due to the introduction of internal frames, since these increase the bending stiffness of the capsule.

**Crew Experience and Vehicle Sizing** Crew experience is an important part of the mission and, if unsuccessfully handled, it could lead to the failure of the mission. The analysis included considerations of factors such as the crew's physical and psychological health, where measures were taken to mitigate any concerns that may concede the well-being of the crew. The major concerns in this mission were protection against radiation, solar flares and micro-gravity.

The effects of radiation and solar flares could be moderated by the careful selection of materials. Such materials would have to additionally be able to withstand the harsh space environment (which faces extreme conditions such as low pressure, thermal cycling and micrometeoroid strikes in addition to the above two), and also be light and cheap to minimise the mass and cost incurred. Hence, composites and light metals such as the Al-Li alloy were carefully chosen for primary structures of the crew capsule and the MAV. Thermal protection systems and corrosion resistant coatings were also investigated for proper functionality of interior structures.

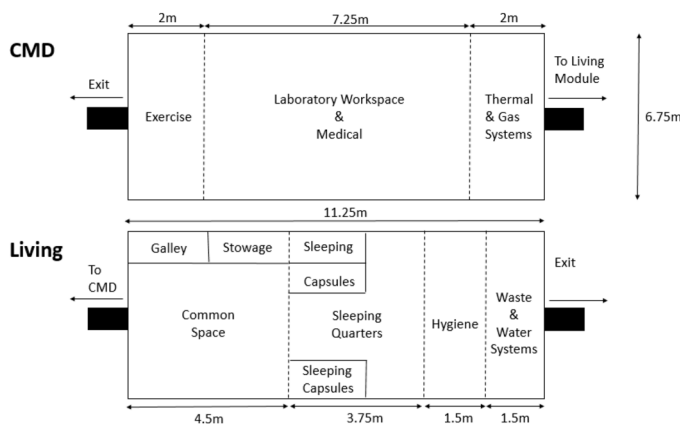
Dimensions of the crew cabin and orbiter were based on research that had already been conducted by NASA.

Once materials and dimensions were confirmed, rough estimates of mass were made by idealising the shape of the MAV and capsule. The orbiter is to be used as the living quarters for the crew during transit and information was extracted from existing space shuttles and altered as necessary to accommodate the requirements of the mission.

## Martian Habitat Design

**Configuration and Location** The location of our Habitat was decided to be 'Elysium Planitia', centered at 3.0°N 154.7°E. This was chosen as it is believed to be abundant with water-ice. Three modules were necessary to ensure mission success. They are the command module, the living module and the greenhouse module.

**Habitat Module** The two habitat modules are formed from a hybrid inflatable structure: a continuous inflatable skin clamped at nodes to an internal 'endoskeleton'. This provides the high volume advantages of an inflatable structure while allowing the mounting of systems on the frame avoiding out of plane loading on the habitat skin. The skin is formed of multiple layers, each with separate functions including strength, gas retention and environmental resistance, etc. The internal frame is formed of SiC reinforced aluminium, cylindrical tubes connected in a pin-jointed manor. Following its construction, the habitat modules will be covered in regolith filled sandbags to provide additional shielding from radiation.



The living and command modules provide a Net Hab-

itable Volume of 315 m<sup>3</sup>, with dimensions measuring 11.25 m × 6.75 m × 2.25 m per module. The functional areas that are included are: Sleeping Quarters, Hygiene, Workspace & Laboratory, Medical, Exercise, Stowage, Dining & Communal Activities, Environmental Control Life Support System (ECLSS) Systems and Airlocks. These are arranged in the configuration as shown above.

**Greenhouse Module** The main purpose of the greenhouse module is to provide enough food for the future colonies to be self-sufficient. Therefore, for the first manned mission, the greenhouse will use as many Martian in-situ resources as possible such as using Martian regolith as growth media. Once the greenhouse proves to be able to grow edible crops in Mars, the greenhouse will be part of the bioregenerative life support systems.

The greenhouse is an 11.2 m long, 3.6 m wide in diameter cylindrical, inflatable structure. ETFE was chosen for the inner and outer structural skin material since it outperformed other transparent materials. The skin will be connected to 8 × Ti-5Al-2.5Sn frames. These materials provide high strength and corrosion resistance which will not degrade for years.

**Systems** The habitat will have the following systems: power generation and regulation, thermal stability control, waste and water management, ISRU, EVA Suits, rovers and communication, and greenhouse maintenance. Power and thermal systems were designed in relatively more detail due to their criticality. The uncertainty of the power systems is minimised by relying upon proven methods in space, namely fission reactors. Radioisotope and solar based systems are also used to test the feasibility of such systems for future colonies, as well as for redundancy.

**In-Situ Resource Utilisation** ISRU was opted to be used as part of the mission to reduce the number of launches to LEO and the landing mass on Mars. ISRU utilises the Martian atmosphere and soil to generate ascent propellant - LOX/LCH<sub>4</sub> as well as oxygen and water for crew life support. With ISRU, all propellant is made in-situ, omitting the need to carry propellant

from Earth and resulting in a landing mass reduction of approximately 40 tonnes.

## Spacecraft Systems and Propulsion

**Earth Launch** For this mission, a launch vehicle with a payload capability of 120 tonnes was required. The reusable two-stage-to-orbit liquid propellant rocket was determined to be the best-suited for our mission. For the propulsion cycle, the staged oxidizer-rich combustion cycle was chosen over the gas generator cycle due to its better thrust performance, efficiency and reusability. For the propellant, liquid oxygen and liquid methane (LOX/LCH<sub>4</sub>) was chosen due to its relative high specific impulse  $I_{sp}$ , lower price and reliability with respect to cooling issues.

The estimated mass of the rocket and propellant was found to be 2336 tonnes and 2121 tonnes respectively. The first stage requires 12 engines with a thrust of 3070 kN each; and the second stage requires 2 engines with a thrust of 2038 kN each. The nozzle was then designed based on the propellant mass flow rate. The diameter of the nozzles were determined to be 1.7 m and 4.5 m for the first and second stages receptively.

**Interplanetary Transfer** After an extensive analysis, the NTP was chosen as the desired propulsive system for interplanetary transport as its engines are capable of specific impulses 3 times that of chemical engines and have thrust of the same order.

Of existing and tested NTP engines, the NERVA ENABLER engine was chosen, with following characteristics:  $I_{sp} = 900$  s,  $F = 450$  kN and  $m = 14000$  kg, assuming future improvements. An iterative analysis was performed to determine the total amount of propellant and number of engines required for each part of the mission.

| Mission                 | T-2 Habitat | T Manned   | T+2 Return |
|-------------------------|-------------|------------|------------|
| Engines                 | 8           | 12         | 12         |
| Mass of LH <sub>2</sub> | 210 tonnes  | 330 tonnes | 94 tonnes  |

**Mars Ascent** The main propulsion system of the MAV will employ the SpaceX Raptor engine (vacuum version) with a specific impulse of 382 s. This will run on LOX/LCH<sub>4</sub> on a full-flow staged combustion cycle. The engine is capable of a maximum thrust of 3500 kN with a throttle capability of 20-100%. Propellant mass required is estimated to be 66 tonnes.

## Aerothermodynamics & EDL Design

**Mars Aerocapture and EDL Concepts** A new entry concept was required to support human landing on Mars, where each landing is expected to carry approximately 20 times the payload mass achievable with existing technology. Aerocapture and hypersonic deceleration with a Mid L/D Rigid Body were adopted primarily for its low developmental risk. Supersonic retro-propulsion was also determined to be critical for supersonic deceleration, while subsonic propulsion would be utilised for the final landing onto Mars surface.

**Entry Vehicle Design** The generic dimensions of the entry vehicle were dictated by its largest indivisible payload, and the constraints of launch rocket fairing. Design optimisation aimed at maximising payload mass fraction, through maximising volumetric efficiency and minimising total heat load. Engineering methods were used to determine the type and thickness of thermal protection system (TPS) required. The placement of the reaction control system and body flaps were deliberated based on aerothermal considerations.

TPS selection was based on mass savings and ability to sustain extreme temperature during aerocapture and EDL. Phenolic Impregnated Carbon Ablator (PICA), a low density and high performance ablator was found to be the best compromise between mass and allowable heat flux.

**Supersonic Retro-Propulsion** Supersonic retro-propulsion (SRP) is a significant hurdle in our plans to reach Mars. The ejection of high enthalpy combustion products into a supersonic opposing free stream can have marked effects on the bow shock, which in

turn can significantly alter aerodynamic characteristics such as drag, stability and control. Higher heating rates also have to be considered.

**Trajectory Shaping** The EDL trajectory was optimized using particle swarm optimization with the primary aim of minimising the propellant mass fraction (PMF). SRP PMFs of the order of 8% partially verified the feasibility of SRP coupled with a rigid mid L/D body.

The dynamic stability and stall characteristics of our re-entry vehicle were also unknowns. Instability of more conventional designs was found to be mainly due to hysteresis in pitching moment characteristics at low supersonic speeds, due to a lagged after-body pressure field. To investigate whether this would also be an issue for our design, a CFD analysis was attempted to gain further understanding of our flow field.

**Mars Descent** The propulsive terminal descent propulsive system is to safely land the landing vehicle from a known initial position and velocity. To complete the manoeuvre with minimal fuel consumption, a fuel optimal control algorithm was solved to obtain the maximum required thrust which, in this case for our landing vehicle of 52 tonnes, is 58 kN. It was also determined that 15 tonnes of LOX/LCH<sub>4</sub> would be required to land the landing vehicle.

The engines were then designed from the thrust required. Each engine will have a mass of 443 kg, a length of 1.4 m, a chamber and nozzle diameter respectively of 0.48 m and 0.53 m, and there will be a total of six engines on each landing vehicle.

## Communications Architecture

A communications system is required for an uninterrupted link with mission control under all circumstances, to support scientific mission goals and provide navigation/telemetry during various mission phases. A two-satellite constellation in Areostationary orbit over the habitat with a 20° phase separation at 4° inclination with ion engines was chosen, providing constant coverage with minimal propellant mass requirements.

Calculations indicated the system would be able to support data rates in excess of 100Mbps.

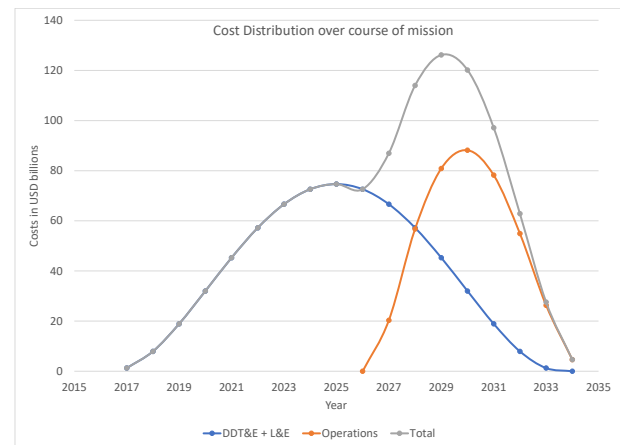
## Cost Estimation

Of the various cost estimation tools available, the Advanced Mission Cost Model was opted for the estimation of the Design, Development, Test and Evaluation (DDT&E) cost due to its relative simplicity and its accounting for most cost drivers.

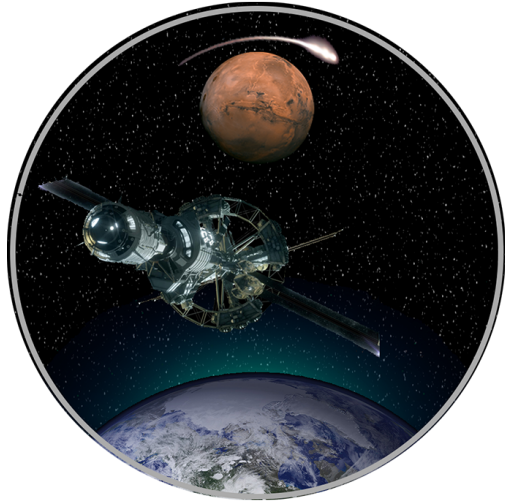
The Launch and Emplacement (L&E) cost was found to be a function of the location factor, while the annual Operations cost was taken to be 10.9% of the total DDT&E cost.

| Type         | Cost                      |
|--------------|---------------------------|
| DDT&E        | US\$597.424 billion       |
| L&E          | US\$81.018 billion        |
| Operations   | US\$410.214 billion       |
| <b>Total</b> | <b>US\$1.089 trillion</b> |

The cost would be split across 18 years, from 2017 - 2034 modelled using a Beta distribution with 50% of the costs being paid in 50% of the duration:



The highest annual cost required would be US\$126 billion in 2029, and although such a cost significantly exceeds NASA's budget, it could be supported with commercial and international collaboration.



# EDEN

THE MANNED MISSION TO MARS

## Aerothermodynamic Design of Mars Entry Vehicle

### Group Design Project Report

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## Abstract

Payload mass associated with a manned mission to Mars far exceeds the capability of the existing Mars entry technology. A new concept will be necessary to facilitate the aim of sending human to Mars under the ambit of Project EDEN. Concepts proposed by research papers will be reviewed to identify feasible solutions. Engineering methods such as the modified newtonian law will be employed to understand the flow characteristics when entering Mars' atmosphere. These knowledge will be used to design and optimise a Mars entry vehicle. A Mid L/D Rigid Body was determined to be the most suitable concept, based primarily on its short minimum time to maturity. Laminar flow was also found to be a good basis for heat flux prediction, enabling the definition of thermal protection system for the various modules in Project EDEN. Methods employed yielded a design that can deliver the payload required for Project EDEN at minimum mass. The final design compares well with design achieved with higher fidelity, and provide the basis for future studies to achieve better optimised production model.



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## Nomenclature

|               |                                                                      |        |                                                                       |
|---------------|----------------------------------------------------------------------|--------|-----------------------------------------------------------------------|
| $\chi$        | Ratio of smaller radius of curvature over larger radius of curvature | $Re_D$ | Reynolds number with reference length $r$                             |
| $\epsilon$    | Local flow angle of attack ( $^\circ$ )                              | $Re_T$ | Transition Reynolds number                                            |
| $\eta_V$      | Volumetric coefficient                                               | $S$    | Surface area ( $m^2$ )                                                |
| $\Lambda$     | $90 - \epsilon$ ( $^\circ$ )                                         | $V$    | Volume ( $m^3$ )                                                      |
| $\rho_1$      | Smaller radius of curvature                                          | $y$    | Height of upper body defined in Figure 4 (m)                          |
| $\rho_2$      | Larger radius of curvature                                           | $y_0$  | Height of vehicle defined in Figure 4 (m)                             |
| $\theta$      | Angle between surface tangent and freestream flow ( $^\circ$ )       | ADEPT  | Adaptable, deployable entry placement technology                      |
| $A$           | Reference area ( $m^2$ )                                             | AETB-8 | Alumina enhanced thermal barrier, with density of $8 \text{ lb/ft}^2$ |
| $C_D$         | Coefficient of drag                                                  | BC     | Ballistic coefficient                                                 |
| $C_H$         | Coefficient of heat transfer                                         | EDL    | Entry, descent and landing                                            |
| $C_L$         | Coefficient of lift                                                  | HIAD   | Hypersonic inflatable aerodynamic decelerator                         |
| $C_P$         | Coefficient of pressure                                              | MAV    | Mars ascent vehicle                                                   |
| $C_{P_{max}}$ | Maximum coefficient of pressure defined in [8, pg.54]                | MRV    | Mid L/D Rigid Vehicle                                                 |
| $L/D$         | Lift-to-drag ratio                                                   | MSL    | Mars Science Laboratory                                               |
| $m$           | Mass (kg)                                                            | PICA   | Phenolic impregnated carbon ablator                                   |
| $M_e$         | Boundary layer edge Mach number                                      | RCS    | Reaction control system                                               |
| $Q_s$         | Stagnation point heat load ( $J/cm^2$ )                              | SIRCA  | Silicone impregnated reusable ceramic ablator                         |
| $q_{c_s}$     | Stagnation point heat flux ( $W/cm^2$ )                              | SRP    | Supersonic retro-propulsion                                           |
| $r$           | Radius of upper fuselage                                             | TPS    | Thermal protection system                                             |

# 1 Introduction

As early as 1964 [1], missions have been flown to Mars to gain greater understanding of it. Information from these missions has revealed the similarities between Mars and Earth, generating interest to eventually colonise Mars. While a manned mission to Mars continues to be elusive, several groups have plans to achieve it in the near future. NASA has a goal of sending humans to Mars in the 2030s [2], while the SpaceX plans to achieve a crewed mission as soon as 2024 [3]. One of the key challenges to overcome though, is the entry, descent and landing (EDL) onto the Mars surface. The Martian atmosphere is too thin (approximately 1% of the Earth's density) to provide substantial deceleration, yet still significant enough to present aerothermal problems during EDL [4]. With the large payload associated with manned missions, the lack of aerodynamic deceleration can lead to extremely high velocity close to Mars surface, as well as limited access to higher elevation landing sites. Large amount of thermal protection system (TPS) will also be required to protect the payload from the aerothermal heating, translating to increased mass and costs, making a manned mission less feasible.

## 1.1 Objectives

The aim of Project EDEN is to achieve human landing on Mars by the end of 2031. To facilitate this goal, the EDL phase must be designed around 2 key objectives: 1. Provide sufficient deceleration for payload associated with a manned mission; 2. Minimise the mass and costs of such a vehicle. Trajectory shaping and entry vehicle design are the main directions in achieving these objectives. This individual report will focus primarily on the design of the Mars entry vehicle. In particular, the existing EDL technology will be reviewed, with future concepts analysed to identify the most feasible solution for Project EDEN. Pressure distribution and flow characteristics in the hypersonic regime around the proposed Mid L/D Rigid Body entry vehicle will be studied to provide predictions of the aerodynamic coefficients and aerothermal heating. These will facilitate the optimisation of the vehicle design and identification of the required TPS. The placement of the reaction control system (RCS)

and the body flaps will also be discussed to complete the vehicle design. The fidelity of the aerodynamic assessments will be addressed, together with future works beyond the proposed final design.

## 1.2 Review of Existing Concepts

Given the thin atmosphere of Mars, the hypersonic portion of the Mars EDL trajectory will continue to lower altitudes relative to an equivalent Earth entry [5]. This leaves little time for the transition to subsonic EDL phase for a safe landing. This also meant that higher elevation landing sites are less accessible. Deceleration at higher altitude can be achieved with a reduction in ballistic coefficient (BC). From Equation 1, a lower ballistic coefficient can be achieved by either decreasing the mass ( $m$ ), or increasing the coefficient of drag ( $C_D$ ) or aerodynamic reference area ( $A$ ) of the entry vehicle.

$$BC = \frac{m}{(C_D)(A)} \quad (1)$$

The Mars Science Laboratory (MSL) mission represents the latest achievement in Mars EDL, successfully delivering a 900 kg payload [5]. To achieve this, an unprecedented 4.6 m diameter aeroshell was adopted to reduce the ballistic coefficient of the entry vehicle, which is still thrice that of the previous missions. A lifting trajectory was also utilised to prolong the descent duration for added deceleration. For a manned mission to Mars, NASA estimated that each landing will need to deliver twenty times the MSL payload [6]. This meant that the reference area of the vehicle will need to be increased by the same magnitude to achieve similar trajectory performance to the MSL. Adopting the same Viking aeroshell design used by the MSL will hence be unfeasible due to huge structural mass involved (vehicle of diameter of 92 m required). A different entry approach will therefore be required to facilitate a manned mission to Mars.

## 2 Conceptualisation

To minimise the amount of propellant required, deceleration from the Mars hyperbolic approach velocity of 7.1 km/s to its orbital velocity of 4.7 km/s [7] can be achieved by using aerodynamic deceleration through the Mars' atmosphere, in a manoeuvre known as aerocapture. The actual EDL, con-

sisting of hypersonic, supersonic and subsonic decelerations, occurs during the actual descent from the orbit to the Mars surface. To facilitate the investigation of aerocapture and EDL concepts, a trajectory model was created using relations highlighted in [8, pg.25], [9], and Mars density model given by NASA [10].

## 2.1 Concepts Identification

From literature review [4], [5], [6], [11], [12], [13], [14], [15], feasible aerocapture and hypersonic decelerator concepts were identified and categorised into flexible heat shield and rigid heat shield concepts. Hypersonic Inflatable Aerodynamic Decelerator (HIAD) is one of the flexible heat shield concepts, making use of inflatable structure to increase the reference area to reduce the ballistic coefficient. This increase in area can also be achieved with deployable structures in the form of a separate concept known as the Adaptable, Deployable Entry and Placement Technology (ADEPT). On the other hand, the rigid heat shield concept, also known as the Mid L/D Rigid Body, accepts the high ballistic coefficient associated with the increase in payload mass. It uses a lifting body to generate increased flight time for prolonged deceleration. Asymmetric HIAD and ADEPT can also provide some of lift, but such a design will complicate the already complex mechanisms in both concepts. Lifting trajectory can instead be achieved with aerodynamic tabs and/ or centre of gravity adjustments in symmetric HIAD and ADEPT.

For supersonic deceleration, the existing method of deploying parachute will not be practical due to the huge size required for the payload involved [5], [15]. The MSL mission for instance requires a parachute diameter of 21 m for less than 1 tonne of payload. Supersonic-retropropulsion (SRP) can be an alternative to supersonic parachute, using retro-rockets to increase the total deceleration of the entry vehicle [16]. However, there are issues associated with rocket initiation under high dynamic pressure as well as flow field unsteadiness due to jet-plume interactions. These will be addressed in detail in a separate paper [17]. For subsonic deceleration, similar powered descent techniques can be employed with less aerodynamic complications.

Details of subsonic deceleration will be discussed by the propulsion team [18]. Subsequent analysis will hence focus on aerocapture and hypersonic deceleration concepts. As HIAD and ADEPT represent similar modes of deceleration, they will be analysed as one entity, with their ballistic and lifting trajectories compared to that of a Mid L/D Rigid Body.

## 2.2 Aerocapture and Hypersonic Decelerator Concepts Analysis

Based on NASA's studies [19], the habitat payload required to support a 6-man manned mission to Mars is approximately 58 tonnes. Literature [4], [14] suggests that the mass of the entry vehicle is approximately 3 times the payload. If sent in a single package, the entry vehicle can potentially weigh up to 200 tonnes. Assuming a fixed design, increasing aerocapture mass decreases altitude margin and can lead to potential impact with the Mars surface (Figure 1).

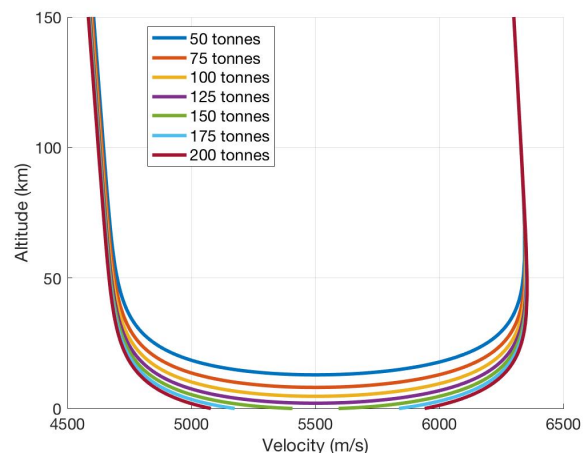


Figure 1: Aerocapture Trajectory with Varying Mass

Larger aerocapture mass also increases the heat load sustained during the aerocapture manoeuvre, increasing the amount of TPS required (Figure 2), and hence overall mass. Therefore, it is proposed that aerocapture be conducted in small packages, to mitigate the above mentioned problems. A nominal payload of 20 tonnes was identified for each package. This will facilitate the delivery of the largest indivisible payload, the Mars Ascent Vehicle (MAV), which weighs 14.68 tonnes [20].

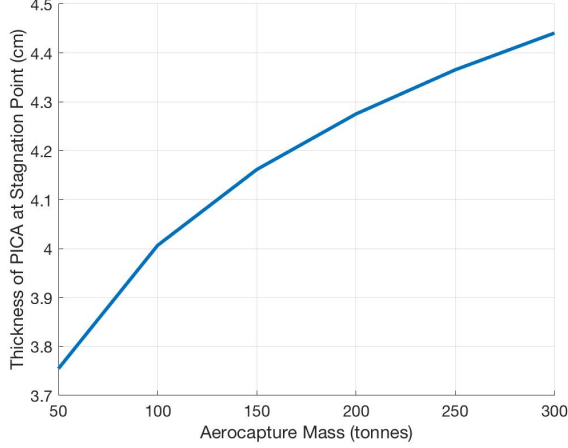


Figure 2: TPS Thickness Required for Aerocapture

Table 1: Parameters for Hypersonic Decelerator Concepts Analysis

|                            | ADEPT/HIAD   | Rigid Body |
|----------------------------|--------------|------------|
| Entry Angle                | 10.8 degrees |            |
| Entry Mass (kg)            | 53,811       | 61,382     |
| L/D                        | 0 / 0.20     | 0.54       |
| Ref Area (m <sup>2</sup> ) | 278          | 65         |
| Nose Radius (m)            | 4.70         | 1.68       |
| BC                         | 121.2        | 377.7      |

To compare the merits of the various hypersonic decelerator concepts, trajectory performances were modeled using the parameters obtained from literature [4], [14] for a common payload of 20 tonnes (Table 1), and the bank profile provided in [14] (See Annex A for detailed results).

ADEPT/HIAD (ballistic/lifting) and Mid L/D Rigid Body concepts are all capable of deceleration to a low altitude Mach number of under 3, the threshold SRP initiation Mach number outlined by NASA [21]. A quick comparison suggests that the ballistic ADEPT/HIAD concept has no apparent advantage in any aspect, streamlining the options to the lifting ADEPT/HIAD and Mid L/D Rigid Body concepts. While the Mid L/D Rigid Body concept registers the lowest Earth equivalent peak g-force, that of the lifting ADEPT/HIAD system is also below the human g-force limit of 4.5. The lifting ADEPT/HIAD concept boasts significantly

lower peak dynamic pressure. Its dynamic pressure at low altitude is also lower, which reduces the complications for SRP initiation. Lifting HIAD/ADEPT concepts also leads to lower peak heat flux and hence smaller amount of TPS required. In fact, if the scalability of the ADEPT/HIAD is not a limitation, the ballistic coefficient of a ADEPT/HIAD system can be reduced significantly. Low altitude subsonic conditions can be achieved, eliminating the need for SRP altogether.

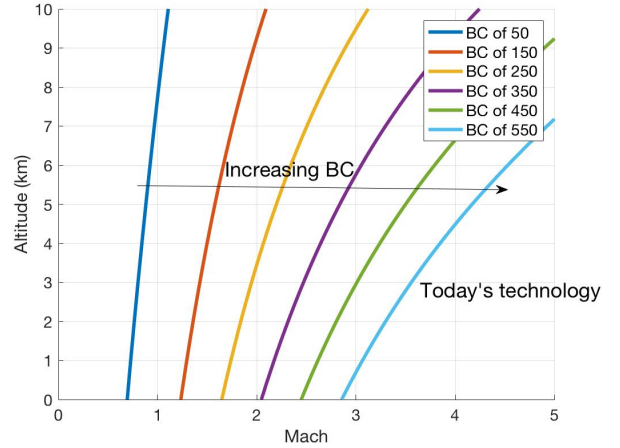


Figure 3: ADEPT/HIAD Trajectory with Varying BC

However, only a 3 m diameter HIAD had been flight tested to date [22], while no known flight test had been conducted for ADEPT. To achieve the parameters highlighted in Table 1, a diameter of 18.8 m must be achieved by 2028, the projected year of launch for the first package [7]. Adopting HIAD/ADEPT presents developmental risk as NASA estimates the minimum time to maturity to be 13 years (as of 2015) [22]. The multitude of moving components in ADEPT/HIAD also raises reliability concerns, especially after prolonged storage during the interplanetary transfer (approx 6-9 months). Fluid interaction with the flexible heat shield of ADEPT/HIAD can further result in varying  $C_L$  and  $C_D$ , increasing the control complexity of such systems. Contrastingly, the Mid L/D Rigid Body leverages on proven materials and manufacturing processes similar to the Space Shuttle Orbiter [14], minimising developmental risks and effort. The payload is also fully protected in the entry vehicle aeroshell, while its higher L/D contributes to greater control authority.

Table 2: Summary of Concepts Comparison

|                                     | Lifting ADEPT/HIAD | Mid L/D Rigid Body | Importance (1-4) |
|-------------------------------------|--------------------|--------------------|------------------|
| Entry Mass (kg)                     | <b>53,811</b>      | 61,382             | 3                |
| Peak G-force                        | 3.5                | <b>2.0</b>         | 2                |
| Dynamic Pressure - SRP (Pa)         | <b>602</b>         | 1453               | 3                |
| Peak Heat Flux (W/cm <sup>2</sup> ) | <b>11</b>          | 28                 | 3                |
| Maturity Time (as of 2015)          | 13 years           | <b>5 years</b>     | 4                |
| Controllability                     | Lower              | <b>Higher</b>      | 3                |
| Reliability                         | Lower              | <b>Higher</b>      | 4                |
| <b>Scores</b>                       | <b>9</b>           | <b>13</b>          | -                |

A summary of the comparisons can be seen in Table 2, where each attribute's influence on the mission completion is quantified (4 being critical). As the maturity and reliability of technology are crucial to the success of the mission, the Mid L/D Rigid Body will be a better option and hence adopted for Project EDEN.

### 2.3 Dimensional Constraints

A generic cylindrical shape was initially decided for the entry vehicle as it maximises volume within the bounds of the cylindrical launch rocket fairing. This also avoids sharp corners, minimising areas of stress concentration, as well as reducing the convective heat flux, which will be discussed in Section 3.2. The maximal dimensions of the entry vehicle is limited by the fairing of the launch rocket, while the lower bounds are dictated by the largest indivisible payload, the MAV. In conjunction with the structure [20] and propulsion [23] teams, it was determined that a cross-section of height 8 m and width of 9 m would accommodate the proposed MAV. It would also fit within the 12 m diameter fairing of the launch rocket. With the internal layout to be defined, a vehicle length of 20 m was adopted from literature [14] for the initial design.

## 3 Aerodynamics Analysis

Predictions of the aerodynamic and aerothermal heating around the proposed vehicle will facilitate the optimisation of the vehicle's shape. This requires an understanding of the pressure distribution and flow characteristics around the body. The

hypersonic regime will be the focus as it is expected to provide the most limiting conditions.

### 3.1 Aerodynamic Coefficients

Anderson [8] suggests that the modified newtonian law (Equation 2) can be used to predict the pressure distributions over generic bodies in the hypersonic regime. The pressure coefficient varies with the angle ( $\theta$ ) between surface tangent and the incoming flow, while  $C_{P_{max}}$  is given in [8, pg.54] as a function of the free-stream Mach number.

$$C_P = C_{P_{max}} \sin^2 \theta \quad (2)$$

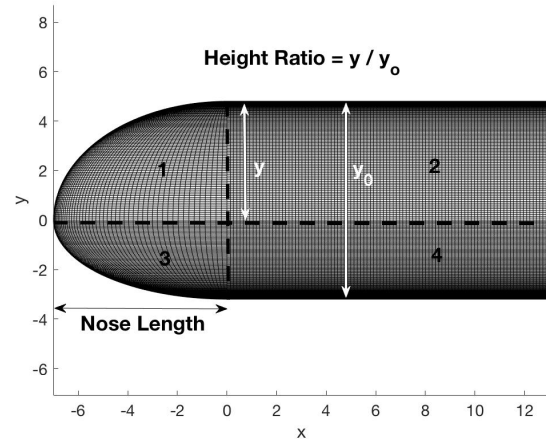


Figure 4: 3D Body Construction

A 3D body was created in Matlab by parameterising the surface using relations highlighted in [24]. To facilitate the optimisation, the 3D body was constructed using 4 segments, allowing the nose length and the height ratio to be varied (Figure



4). The pressure coefficient ( $C_P$ ) at each discretised surface can be computed and integrated over the entire surface to find the lift, drag and pitching moment coefficients.

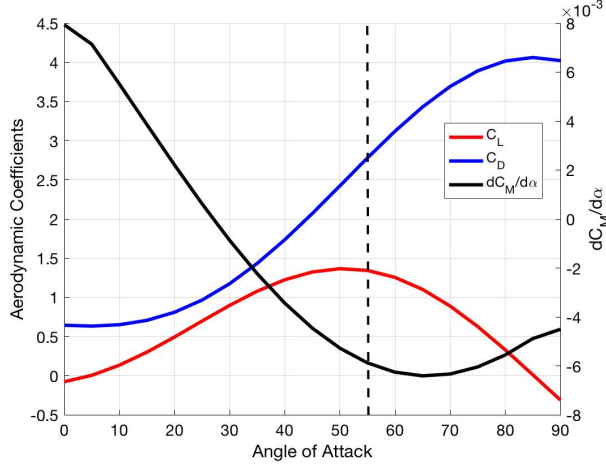


Figure 5: Aerodynamic Coefficients and Longitudinal Static Stability

To simplify the trajectory profile, the vehicle's angle of attack was kept constant. An ideal angle of  $55^\circ$  was determined using a generic shape, based on a compromise of maximum lift, drag and longitudinal static stability (Figure 7). This was found to be applicable even when the shape is varied. For instance, the vehicle remains longitudinally stable at  $55^\circ$  for varying nose lengths (Figure 6).

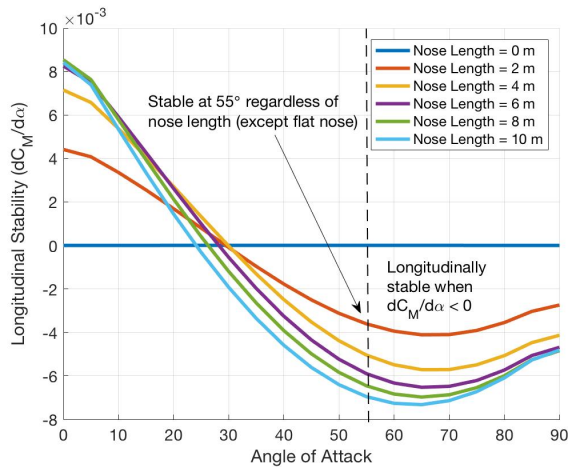


Figure 6: Longitudinal Static Stability with Varying Nose Length

Agreement with the CFD results presented in [14] for similar shapes provides confidence in the fi-

delity of the present method. However, extension of the method to low supersonic conditions should be avoided, particularly due to the assumption that the leeward pressure is the same as that of the free-stream ( $C_P = 0$ ). Flow expansion on the leeward surface can instead result in negative  $C_P$  in low Mach flow.

### 3.2 Aerothermal Predictions

Aerothermal predictions are typically achieved with CFD modelling due to the complex nature of both hypersonic flow and high temperature gas dynamics. Non-zero angle of attack as well as surface curvature of 3D body further complicates the streamlines distribution. However, 3D CFD modelling can be time consuming and hence impractical for initial design. Instead, engineering methods from [25], [26] are utilised to predict the heat transfer distribution around a 3D body. The heat transfer coefficient ( $C_H$ ) at any surface on the windward side is given by Equation 3. Subscript 0 denotes the contributions due to flow on a flat plate, while 1 represents that of stagnation heating on a blunt body.

$$C_H = C_{H_0} + C_{H_1} \sin\theta \quad (3)$$

For a 3D body, its stagnation point heat transfer coefficient ( $C_{H_{3D}}$ ) can be obtained from the 2D stagnation point heat transfer coefficient ( $C_{H_{2D}}$ ) of an equivalent sphere as per Equation 4. On each surface, two radii of curvature exist, defined by the coordinate system in [26].  $\rho_1$  and  $\rho_2$  represents the smaller and larger radius of curvature respectively. The equivalent sphere adopts  $\rho_1$  as its radius for 2D stagnation point calculation.

$$C_{H_{3D}} = \sqrt{1 + \chi} \times C_{H_{2D}} \quad (4)$$

$$\chi = \rho_1 / \rho_2 \quad (5)$$

Convective stagnation heat transfer is computed using Sutton and Graves correlations [27], while the radiative stagnation heat transfer expression can be found in [28]. Of note, radiative heat transfer is only expected above 6 km/s for Mars and 9 km/s for Earth, therefore likely to only affect aerocapture manoeuvres. Reference temperature method [8, pg.286] is used to find the heat transfer coefficient over a flat plate, with different treatments for laminar and turbulent flows.

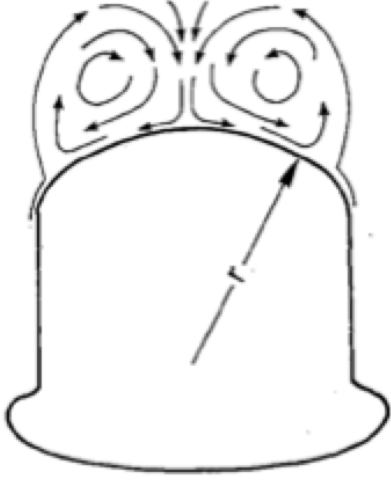


Figure 7: Cross Section of Shuttle's Fuselage with Leeward Vortex Flow [29]

Heat transfer on the leeward region is less straightforward as it is dominated by vortex flow. Empirical method [29] has been used to predict the center-line heating ( $q_c$ ) on the Space Shuttle Orbiter fuselage and is deemed applicable for the present design due to the similarity in shape (slender leeward surface). The radius of the upper fuselage ( $r$ ) is used to compute the Reynolds number ( $Re_D$ ) and stagnation point heating ( $q_{cs}$ ), together with the local properties of leeward flowfield.  $\Lambda = 90deg - \epsilon$ , where  $\epsilon$  is the local flow angle of attack. Since  $\epsilon$  can only be determined in a wind tunnel test, a conservative approach will be to assume  $\Lambda = 0$  for maximum possible heating.

$$q_c = 7.5q_{cs} Re_D^{0.3} (2.975 + 3.428 \cos \Lambda) \times 10^{-4} \quad (6)$$

### 3.3 Flow Transition

Figure 8 shows how turbulent flow on the proposed Mid L/D Rigid Body can result in significant increase in heat flux, and hence the importance of transition prediction. Equation 7 predicts the transition Reynolds number ( $Re_T$ ) as a function of the boundary layer edge Mach number ( $M_e$ ).

$$\log_{10}(Re_T) = 6.421 \exp[1.209 \times 10^{-4} M_e^{2.641}] \quad (7)$$

Figure 9 suggests that the local Reynolds number at the end of the body of the proposed concept is only expected to exceed the transition Reynolds number at 23 km above surface during EDL. Since the peak flux is expected to occur at a much higher altitude of 35 km, the assumption of laminar flow

can be applied for the design of the vehicle during EDL. Laminar flow was also predicted when both Mars and Earth aerocaptures. Subsequent shape optimisation is expected to have negligible impact on the flow transition, as the body curvature's influence on transition is only significant if the ratio of the boundary layer thickness to the radius of curvature of body is more than unity [30].

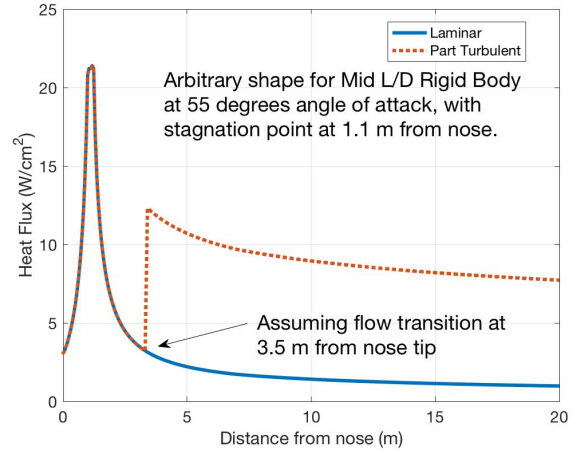


Figure 8: Effect of Turbulent Flow on Heat Flux

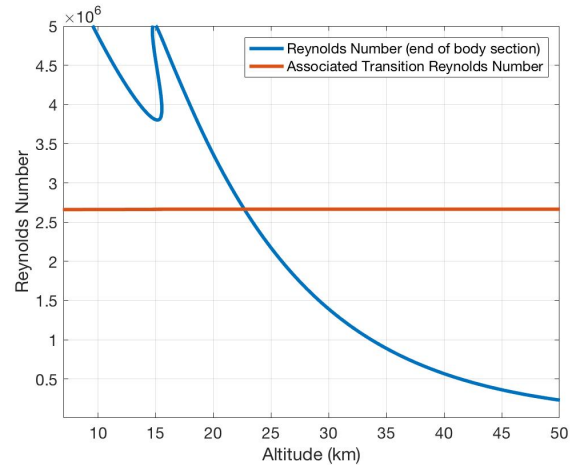


Figure 9: End of Body Transition Prediction for Proposed Concept at 55 Degrees during EDL

### 3.4 Entropy Swallowing

To predict the flat plate heating at each surface, the boundary layer edge conditions are used. This is obtained assuming the streamline on the edge of the boundary layer is the same streamline that has passed through the normal shock (i.e normal

shock entropy). However, boundary layer thickness grows with distance from the stagnation point. The boundary layer swallows streamlines of lower entropies (Figure 10), leading to lower boundary layer edge entropy. This results in a slight increase in heat flux towards the end of the body (Figure 11). Prediction of entropy swallowing requires complex definition of boundary layer growth and shock shape. Instead, the boundary layer edge entropy is assumed to be the same as the local bow shock entropy. The bow shock is in turn assumed to have the same contour as the body surface. This ensures that the heat flux approximated is higher than reality, avoiding an under-designed TPS.

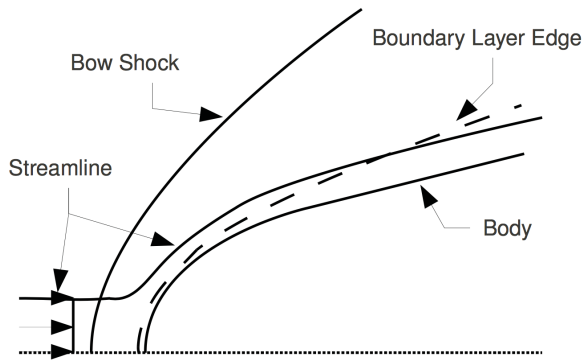


Figure 10: Swallowing of Lower Entropy Streamline with Boundary Layer Growth [31]

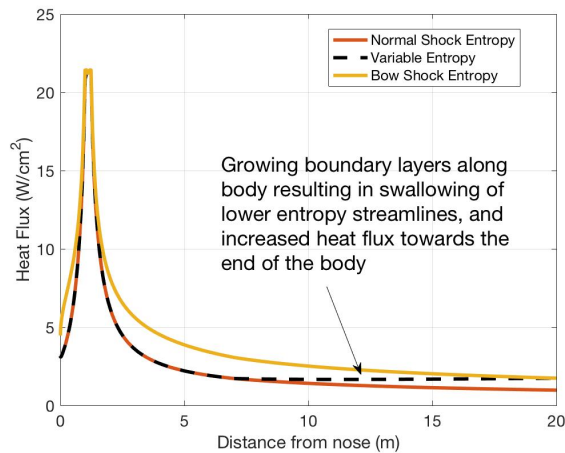


Figure 11: Effect of Entropy on Heat Flux

Flow transition is further delayed by the consideration of entropy swallowing (Figure 12), due to the higher transition Reynolds number created by the lower entropy boundary layer edge conditions.

This further vindicates the assumption of laminar flow for heating approximation.

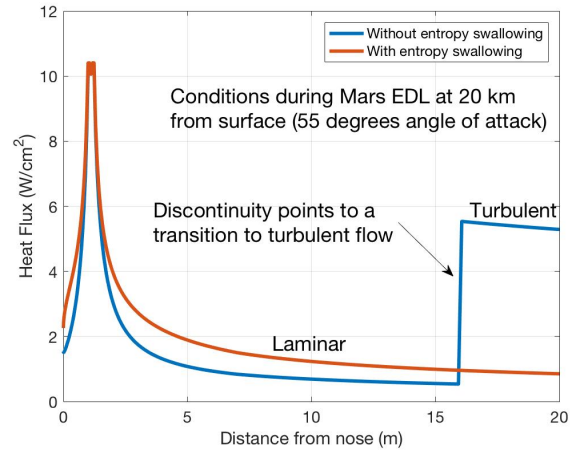


Figure 12: Effect of Entropy Swallowing on Flow Transition

### 3.5 Uncertainty Factor

Methods discussed thus far do not take into consideration surface roughness and surface catalysis. Studies [32] have shown that such sensitivities can result in uncertainties as high as 41 %. Drawing reference from the design of the MSL entry capsule [33], an uncertainty factor of 1.5 will be applied to the computed heat load when determining the thickness of TPS required.

## 4 Shape Optimisation

A generic cylindrical shape is first chosen to minimise sharp corners, which promotes stress concentrations and high stagnation heat flux. It also enables the vehicle to fit within the cylindrical launch rocket fairing. The shape can be optimised by varying the nose length and the height ratio defined in Section 3.1, to maximise the payload mass fraction. One of the optimisation objectives would hence be maximising the enclosed volume ( $V$ ), while minimising the structural mass. This can be subsumed under volumetric efficiency ( $\eta_V$ ), a dimensionless term given in [34], where  $S$  is the surface area.

$$\eta_V = 6\sqrt{\pi} \frac{V}{S^{3/2}} \quad (8)$$

Mass also manifests in the form of TPS, which can be reduced by keeping the heat load accumulated

Table 3: Summary of Attributes of Shortlisted Shapes

| Shape                              | 1       | 2       | 3       | 4       |
|------------------------------------|---------|---------|---------|---------|
| Height Ratio                       | 0.5     |         | 0.6     |         |
| Nose Length (m)                    | 7       | 9       | 5       | 7       |
| Enclosed Volume (m <sup>3</sup> )  | 999.03  | 961.33  | 1036.73 | 999.03  |
| Aeroshell Weight (kg)              | 2887.96 | 2816.12 | 2973.56 | 2894.74 |
| Heatshield Weight (kg)             | 3301.28 | 3180.90 | 3307.66 | 3164.79 |
| Aeroshell + Heatshield Weight (kg) | 6189.24 | 5997.01 | 6281.23 | 6059.52 |
| Peak Dynamic Pressure (kPa)        | 13.68   | 13.72   | 12.73   | 12.73   |

during the trajectory to a minimum. As a reference, stagnation point heat load ( $Q_s$ ) will be used. A merit function of  $\frac{\eta V}{Q_s}$  should hence be maximised.

A merit function density plot (Figure 13) was obtained by putting different shapes through the optimised EDL trajectory provided by [17]. The EDL entry mass was assumed to be 60 tonnes, based on estimates provided in [14] for 20 tonnes of payload.

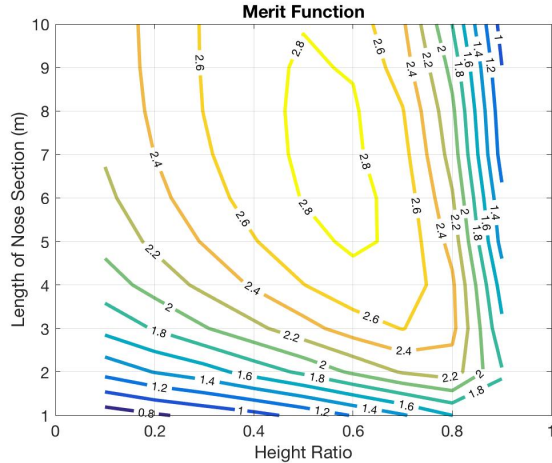


Figure 13: Merit Function Density Plot for Varying Shapes

4 shapes with the highest merit function value were shortlisted, with their attributes compared in Table 3. The aeroshell weight is obtained assuming an areal density of 5 kg/m<sup>2</sup>. The heatshield weight is determined by applying varying thickness of PICA, a form of TPS, based on the heat load on each windward surface of the 3D body. Generally, it can be seen that increasing nose length decreases the heatshield weight, while a larger height ratio leads to lower peak dynamic pressure. In conjunction with the structure team [20], it was determined that a height ratio of 0.6 would better accommo-

date the MAV. Between shape 3 and 4, the latter represents better mass saving and hence chosen for the proposed entry vehicle.

## 5 Thermal Protection System

In conjunction with Kandangwa [35] and the structure team [36], the TPS materials for the proposed entry vehicle were determined. For the windward surface, an ablative material (PICA) was selected due to the high heat flux expected. Ablation converts the surface materials into gaseous products within the boundary layer, and drastically reduces the convective and radiative heating that reaches the surface. On the leeward side, the heat flux is too low for ablation and an insulator such as Alumina Enhanced Thermal Barrier (AETB-8) would be more suitable. Table 4 highlights the conditions for aerocapture and EDL onto Mars by the habitat module, provided by the trajectory [7] and the project management [37] teams. Since the same vehicle design will be employed for other modules involved in Earth aerocapture, the Earth aerocapture by the orbiter module will also be investigated.

Table 4: Summary of Trajectories Parameters

|                       | EDL  | Aerocapture |       |
|-----------------------|------|-------------|-------|
| Atmosphere            | Mars | Mars        | Earth |
| Entry Mass (tonnes)   | 72   | 89          | 54    |
| Entry Velocity (km/s) | 4.7  | 7.1         | 10.7  |
| Entry Altitude (km)   | 250  | 150         | 250   |
| Exit Velocity (km/s)  | N.A. | 4.7         | 7.7   |

### 5.1 Windward

For EDL Mars, the peak flux occurred at the tip of the nose, instead of the stagnation point located



1.11 m aft (Figure 14). The higher flux at the tip of the nose can be attributed to the smaller local radius of curvature (higher curvature), which is inversely proportional to the magnitude of convective heat flux. Higher curvature also accelerates the local flow, resulting in higher flat plate heating.

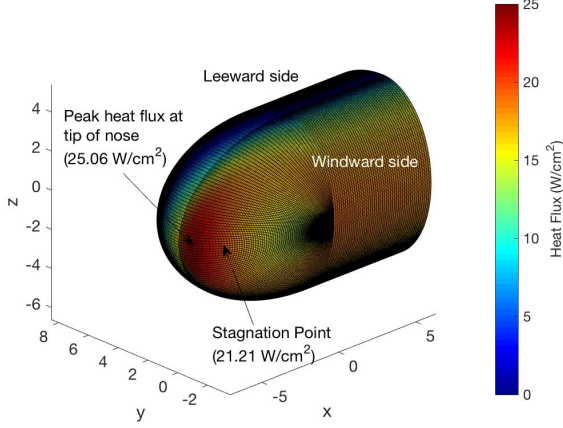


Figure 14: Windward Heat Flux Distribution for Mars EDL (Peak Flux Conditions)

However, for both aerocapture scenarios, the region of peak flux coincides with the stagnation point (Figure 16). Due to the higher velocities during aerocapture, radiative heating is no longer negligible (Table 5). As radiative heating varies proportionally with the radius of curvature, higher radiative heating occurs away from the tip of the nose, contributing to the shift in location of the peak heat flux.

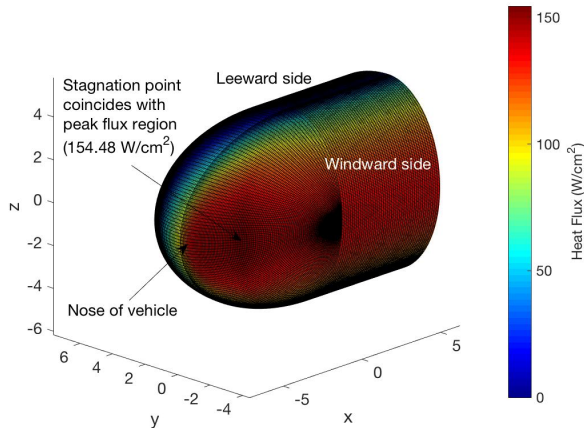


Figure 15: Windward Heat Flux Distribution for Mars Aerocapture (Peak Flux Conditions)

Integrating the heat flux over the duration of the each trajectory gives the total heat load, which is used to compute the thickness required at each surface using the correlations provided in [38]. Due to the manufacturing limitation of PICA, a minimum thickness of 3.2 cm is imposed regardless of heat load. However, a PICA heat shield is still lighter compared to alternatives. A PICA heat shield for Mars EDL weighs 3164 kg, compared to 4694 kg when SIRCA, an alternate form of TPS, is used.

Table 5: Windward Peak Heat Flux Conditions

|                                      | EDL   | Aerocapture |        |
|--------------------------------------|-------|-------------|--------|
| Atmosphere                           | Mars  | Mars        | Earth  |
| Altitude (km)                        | 33100 | 31311       | 64227  |
| Velocity (km/s)                      | 4214  | 6455        | 9827   |
| Peak Flux (W/cm <sup>2</sup> )       | 25.06 | 154.48      | 167.05 |
| Stagnation Flux (W/cm <sup>2</sup> ) | 21.21 | 153.49      | 166.25 |
| % Radiative Flux                     | 0.0   | 47.20       | 36.89  |

## 5.2 Leeward

Table 6: Leeward Heating Rates at Peak Heat Flux Conditions

|                                   | EDL   | Aerocapture |       |
|-----------------------------------|-------|-------------|-------|
| Atmosphere                        | Mars  | Mars        | Earth |
| Leeward Flux (W/cm <sup>2</sup> ) | 3.25  | 11.52       | 4.13  |
| % of Peak Heat Flux               | 12.97 | 7.46        | 2.47  |

Heat flux observed on the leeward side is significantly lower than the windward side (Table 6). However, the temperature on the leeward surface can still be too high for the core structure. Using Stefan-Boltzman's Law, the radiative equilibrium temperature of the body was found from the predicted heat flux. Even during the relatively less severe conditions of Mars EDL, the leeward temperature is 893 K, close to the melting point of common structural materials such as aluminum. In conjunction with the structure team [36], it was determined that the insulation tiles AETB-8 would provide the necessary thermal protection for even the worst scenario (aerocapture Mars), with the thickness determined from [40].

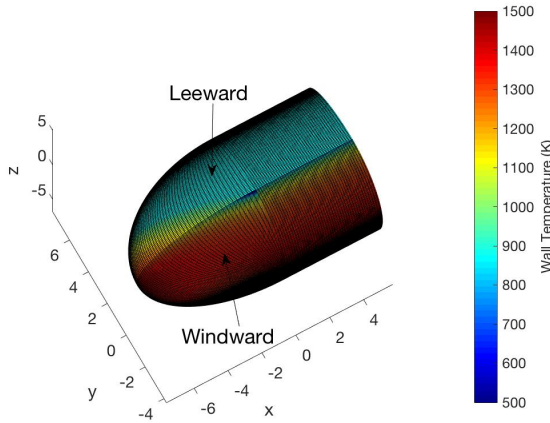


Figure 16: Temperature Distribution for Mars EDL (Peak Flux Conditions)

### 5.3 TPS Configurations

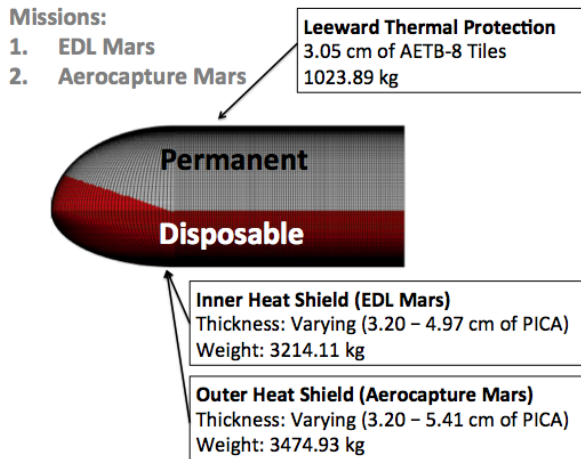


Figure 17: TPS Configuration for Habitat Module

Ablation of the heat shield during the high heat flux conditions removes some of the surface material, resulting in uneven surface and increased roughness. This can lead to early flow transition,

resulting in augmented turbulent heating. Heat shield will therefore need to be shed after each aerodynamic manoeuvre (e.g. aerocapture into Mars) to avoid augmented heating in the next phase. For instance, vehicles designed for Mars landing need to have separated heat shields for aerocapture into Mars and Mars EDL respectively (Figure 17). This applies to other modules involved in multiple manoeuvres. As the leeward TPS is non-ablative, it can be attached to the vehicle permanently, with thickness determined by the worst leeward conditions. Summary of the TPS configurations for the various modules can be found in Table 7.

## 6 External Features

### 6.1 Reaction Control System

Both aerocapture and EDL trajectories are shaped by changing the lift vector of the body using bank profiles. To achieve the desired bank at each time step, reaction control system (RCS) is required. RCS produces under-expanded jets, which can interact with the flow around the vehicle to result in undesirable pressure and heating alterations. Studies [41] suggest that RCS jets aimed into the oncoming flow will result in horseshoe shock, which augments local heating upstream of the RCS position. Furthermore, RCS openings will promote transition to turbulent flow, which further augment heating. Positioning the RCS on the windward side will therefore require additional TPS.

Positioning the RCS on the leeward side is not without complications. If the jet is sufficiently close to the mixing layer separating the supersonic flow and the recirculating region, it can penetrate the layer and force the higher enthalpy supersonic flow close to the wall, resulting in augmented heat-

Table 7: Summary of TPS Configurations for Various Modules

|                       | Habitat Module |         | Orbiter Module |             | Propellant Module |
|-----------------------|----------------|---------|----------------|-------------|-------------------|
| Manoeuvre             | Aerocapture    | EDL     | Aerocapture    | Aerocapture | Aerocapture       |
| Atmosphere            | Mars           | Mars    | Mars           | Earth       | Mars              |
| Entry Mass (tonnes)   | 89             | 72      | 57             | 54          | 113               |
| Heat Shield Mass (kg) | 3473.93        | 3214.11 | 3271.60        | 3837.69     | 3595.51           |
| Leeward TPS Mass (kg) | 1023.89        |         | 895.91         |             | 1087.88           |

ing and pressure [42]. Hence, the ideal placement of the RCS should be towards the end of the body on the leeward surface, where its impact on the rest of the body is likely to be minimal. More precise placement of the RCS will also depend on the control authority required of the system.

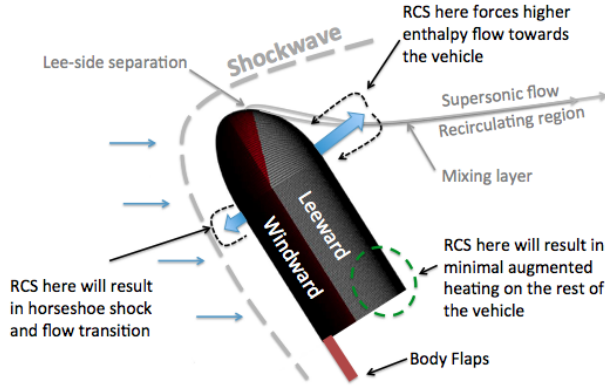


Figure 18: Effects of RCS Placement on Vehicle

## 6.2 Body Flaps

To reduce the negative impact of RCS, body flap will be introduced to minimise the usage of RCS during each manoeuvre. In particular, the body flaps will be responsible for maintaining the vehicle in the desired angle of attack. To be effective, the flaps need to be positioned on the windward side, where the coefficient of pressure is non-zero. Applying the modified newtonian law, the inclination of the flaps relative to the oncoming flow will determine the control force generated by the flaps and hence the moment contributed. However, the flaps will be subjected to extremely high heat flux and temperature on the windward side ( $151.39 \text{ W/cm}^2$  and  $2334 \text{ K}$  during an aerocapture into Earth's orbit). Similar to the windward heat shield, multiple layers will be necessary if ablative TPS is utilised. Instead, the mass of the body flaps can be minimised by adopting heat-directed carbon-carbon composite architecture. The composite forms the core structure without additional TPS attached. Heat transferred to the flaps can be directed efficiently towards the leeward side [43], allowing the flap structure to sustain high heat flux for long duration.

## 7 Discussions

### 7.1 Final Design

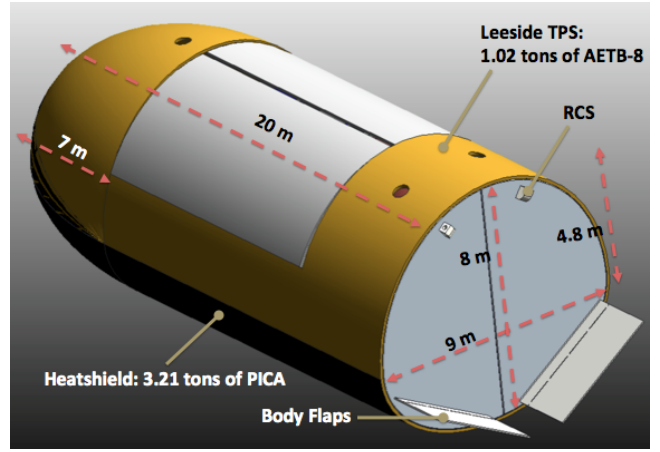


Figure 19: Final Design of Mars Entry Vehicle (Habitat Module Configuration for EDL Mars)

Figure 19 shows the final design of the Mars entry vehicle, with a nose length of 7 m and height ratio of 0.6. This design fulfills the requirement of facilitating a manned mission to Mars, by supporting the delivery of the largest indivisible payload required, the MAV (See Annex C for more details). Methods used also serve to minimise the payload mass fraction, by increasing the volumetric efficiency and reducing the heat load and hence TPS mass. A sanity check can be carried out on the present design by comparing it to a separate design for the Mid L/D Rigid Body concept, proposed in [14].

Table 8: Comparison between Final Design and Cobra MRV for Mars EDL

|                                 | EDEN  | Cobra MRV |
|---------------------------------|-------|-----------|
| Entry Mass (tonnes)             | 70    | 60        |
| Entry Flightpath ( $^{\circ}$ ) | -15.0 | -10.8     |
| Entry Velocity (km/s)           | 4.7   | 4.7       |
| $C_D$                           | 2.78  | 2.50      |
| L/D                             | 0.48  | 0.54      |
| TPS Mass Fraction (%)           | 6.05  | 4.62      |

Table 8 shows that the characteristics of the final design are reasonably similar to a design achieved with higher fidelity techniques such as CFD [14], validating the methods employed used. The higher

TPS mass fraction obtained can be partly attributed to the conservative approach adopted, such as using a uniform maximum heating rate on the leeward side, as well as assuming that the boundary layer edge conditions adopts the local bow shock entropy.

## 7.2 Future Works

One key assumption made at the beginning of the design process is the vehicle length of 20 m. The assumed length is able to accommodate the largest indivisible payload comfortably, but might have resulted in an over-designed vehicle. Future works would better define the volume required by essential systems such as avionics and propellant tanks, allowing the length and hence mass be further optimised.

Current optimisation process also utilised an optimised trajectory based on initial estimates of the vehicle characteristics. To achieve greater accuracy in the vehicle design, the vehicle characteristics of the current design (such as mass, coefficient of drag) should be provided as inputs to generate a new optimised trajectory. This will therefore be an iterative process, and should be necessarily continued until the obtained numbers converge. Iteration will also need to be conducted in conjunction with the structure team to ensure the structural efficiency for an optimised overall mass.

Present work optimises the design based on the requirements of a module conducting aerocapture into Mars orbit, followed by EDL onto Mars. Modules such as the orbiters do not undergo EDL Mars, but instead need to do an additional aerocapture into Earth's orbit. The greater contribution of radiative heat flux to the total heating (Table 5) meant that the optimised shape for these scenarios would likely be different. While utilising a single vehicle design limits the developmental costs and effort, investigation can be carried to see if the merits of a bespoke design for each module is more rewarding. Similarly, additional investigation should be conducted to determine if a different TPS configuration will be required for modules that conduct aerocapture followed by an immediate EDL onto Mars without orbiting.

Finally, it must be cautioned that the present design should only be used at the conceptual stage of the design cycle. Present design used a correlation to determine that the flow along the vehicle's windward surface is laminar. Empirical method was also applied for the leeward heating, given the complexity of leeward vortex flow. To validate the applicability of these assumptions/methods, CFD analysis as well as actual wind tunnel testing will need to be conducted on the present design. Effects of ablation on transition as well as boundary layer as a whole have also been largely overlooked and need to be further studied to provide a more complete understanding of the flow around the present design, optimising it further if necessary.

## 8 Conclusions

The existing Mars entry technology has been reviewed and deemed to be unfeasible for supporting the large payload mass associated with a manned mission to Mars. Alternative concepts were studied with results summarised in Table 2. It was determined that a Mid L/D Rigid Body concept would be the most viable option primarily due to its short minimum time to maturity and high reliability. Engineering methods were employed in place of CFD to provide reasonable predictions of the flow characteristics during aerocapture and EDL. Laminar aerothermal heating is expected during each of the investigated manoeuvres and these knowledge were used to optimise the vehicle design. Due to the different mass and manoeuvre undertaken by each of the modules involved in Project EDEN, different TPS configuration is required, summarised in Table 7. The final design was achieved after considering the aerothermal impact of RCS and body flaps placements. It compares well with design achieved with higher fidelity methods, validating the design techniques. This design enables the delivery of the largest indivisible payload required for Project EDEN, while achieving minimum mass within the limitations of the methods used. Future works will include: iterations in conjunction with trajectory shaping and structural analysis; bespoke shape optimisation for each module type; CFD / wind tunnel studies on flow transition, leeward flow characteristics and ablation influence on boundary layer.



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## A Concepts Comparison

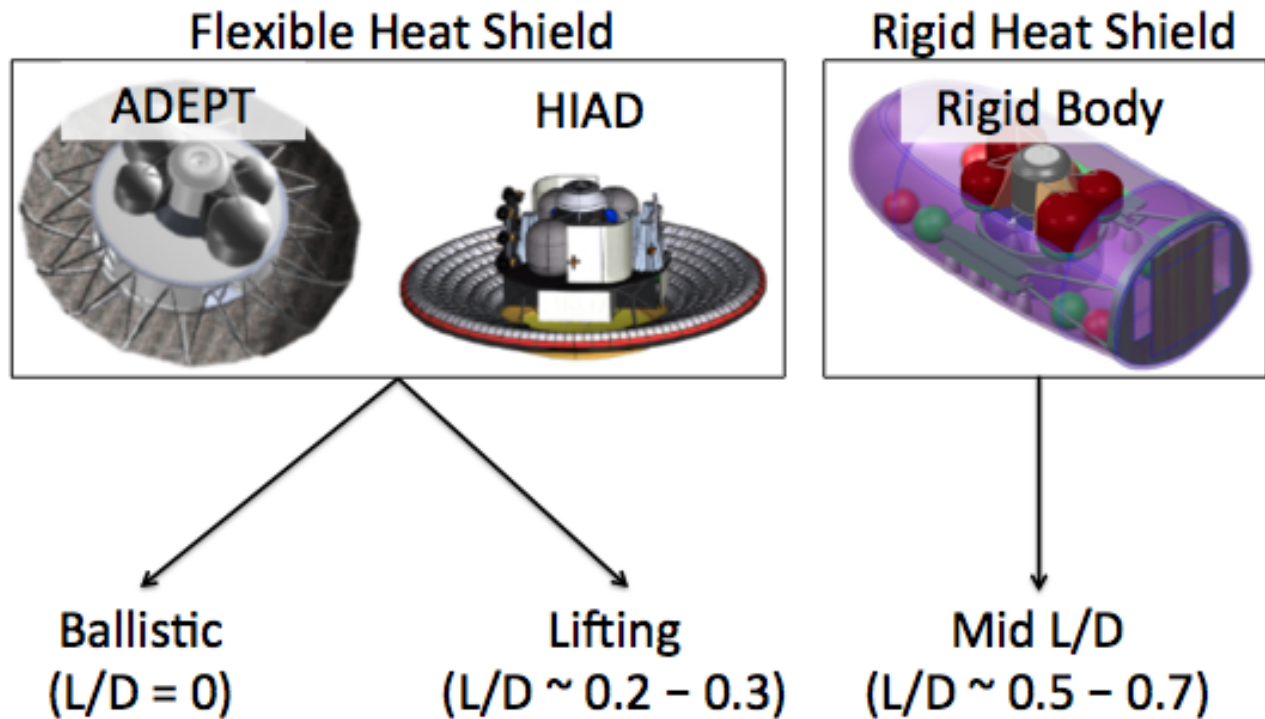


Figure 20: Investigated Concepts for Aerocapture and Hypersonic Deceleration

Figure 21 to 24 show the results of trajectory comparison among the ballistic ADEPT/HIAD, lifting ADEPT/HIAD and Mid L/D Rigid Body concepts. In all aspects, the ballistic ADEPT/HIAD concept has no appreciable advantage over its competitors. While the Mid L/D Rigid Body can attain a lower peak g-force during its trajectory, the lifting ADEPT/HIAD concept has a peak g-force of 3, which is still under the human limit of 4.5. On the other hand, the peak heat flux and peak dynamic pressure of the lifting ADEPT/HIAD concept is significantly lower than that of the Mid L/D Rigid Body. On the basis of this comparison, the lifting ADEPT/HIAD would appear to be the best option among the concepts investigated. However, the attainment of the modeled performance contingent upon the ADEPT/HIAD's ability to be scaled to a diameter of 18.8 m.

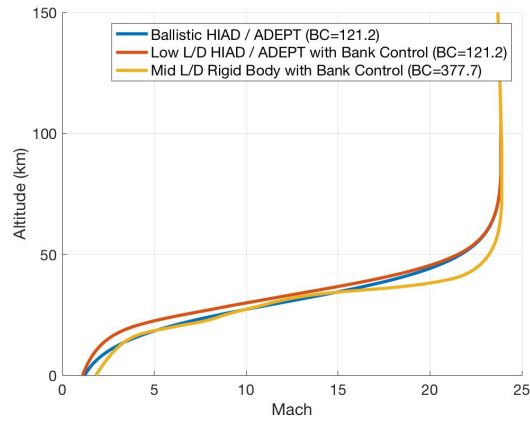


Figure 21: Plot of Altitude against Mach Number

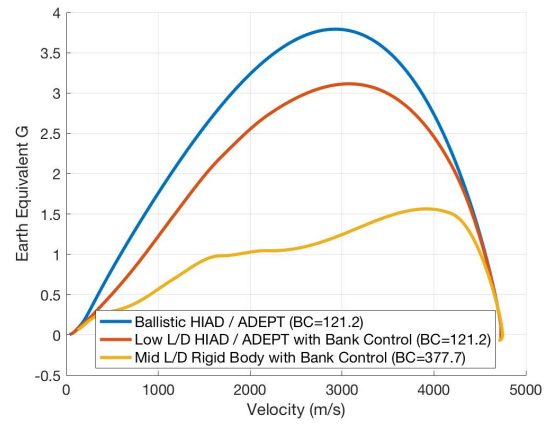


Figure 22: Plot of G-force against Velocity

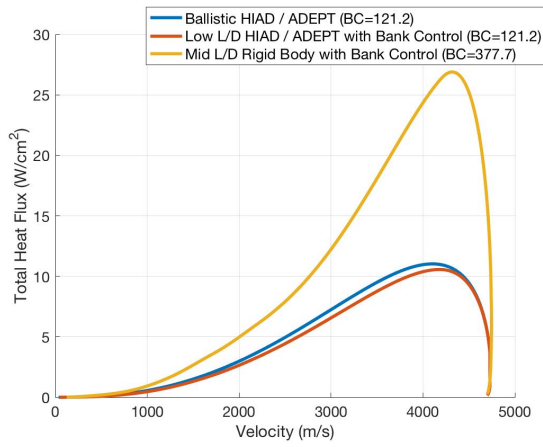


Figure 23: Plot of Heat Flux against Velocity

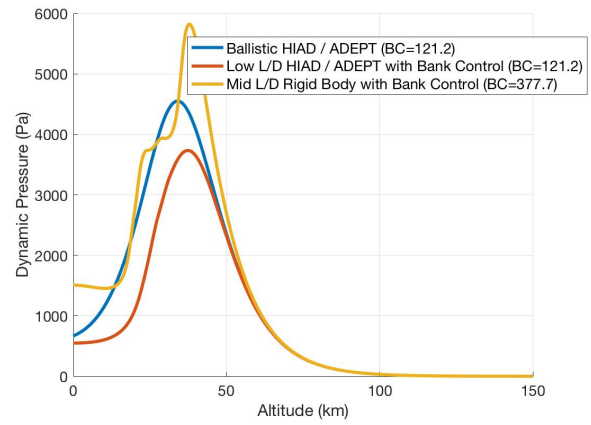


Figure 24: Plot of Dynamic Pressure against Altitude

## B Shape Optimisation Results

Figure 25 to 31 present how the change in nose length and height ratio can influence specific characteristics of the Mars entry vehicle. Similar to varying the nose length, varying height ratio has no drastic effect on the longitudinal stability of the vehicle (Figure 32).

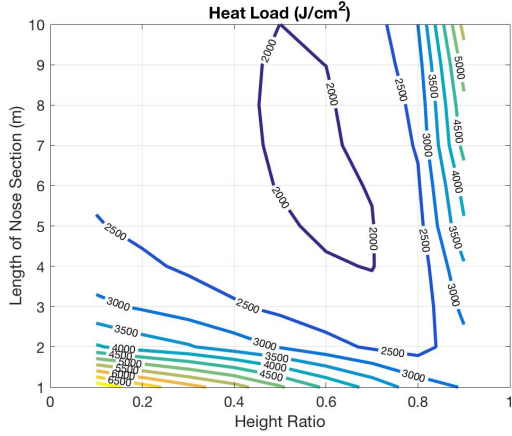


Figure 25: Effect of Varying Shape on Stagnation Heat Load

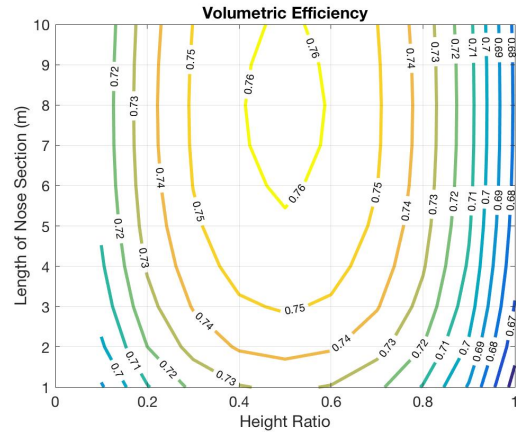


Figure 26: Effect of Varying Shape on Volumetric Efficiency

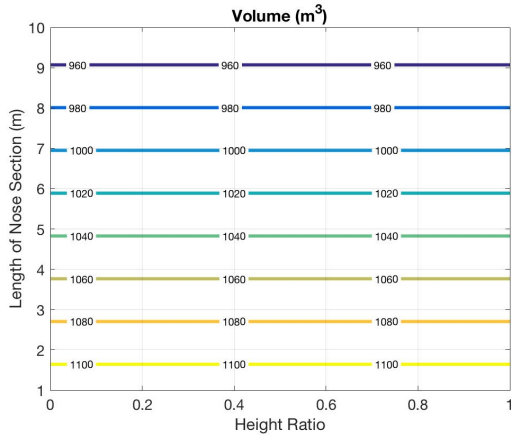


Figure 27: Effect of Varying Shape on Enclosed Volume

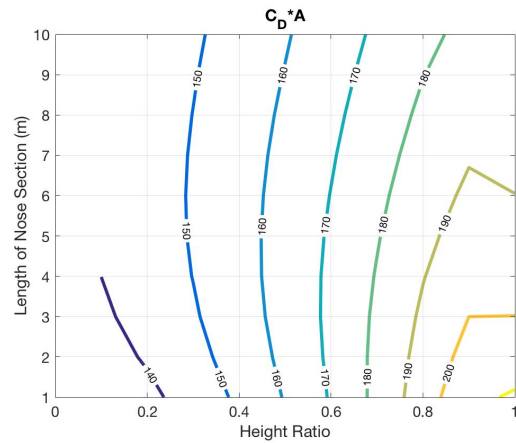


Figure 28: Effect of Varying Shape on  $C_D \times A$

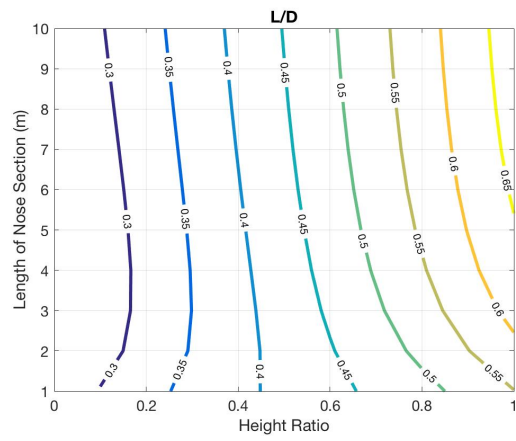


Figure 29: Effect of Varying Shape on Lift to Drag Ratio

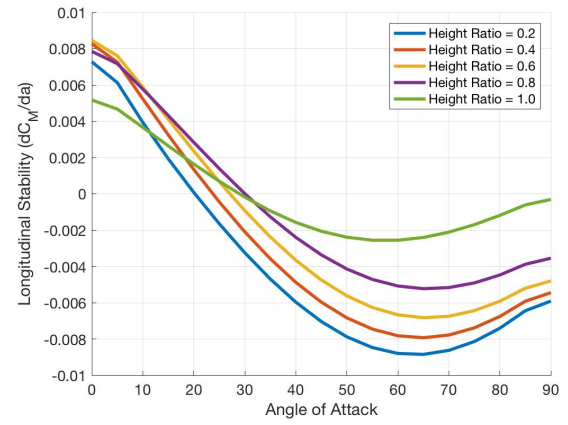


Figure 30: Effect of Height Ratio on Longitudinal Stability



## C Details of Final Design

Table 9: Summary of Dimensions and Mass of Various Modules

|                                      | Habitat Module | Orbiter Module | Propellant Module |
|--------------------------------------|----------------|----------------|-------------------|
| Length (m)                           | 20             | 20             | 20                |
| Height (m)                           | 8              | 8              | 8                 |
| Width (m)                            | 9              | 9              | 9                 |
| Nose Length (m)                      | 7              | 7              | 7                 |
| Height Ratio                         | 0.6            | 0.6            | 0.6               |
| Approximate Structural Mass (tonnes) | 16             | 16             | 16                |
| Inner Heat Shield Mass (tonnes)      | 3.21           | 3.84           | 3.60              |
| Outer Heat Shield Mass (tonnes)      | 3.47           | 3.27           | N.A.              |
| Leeward TPS Mass (tonnes)            | 1.02           | 0.89           | 1.09              |
| Total Aeroshell Mass (tonnes)        | 23.70          | 24.00          | 20.69             |

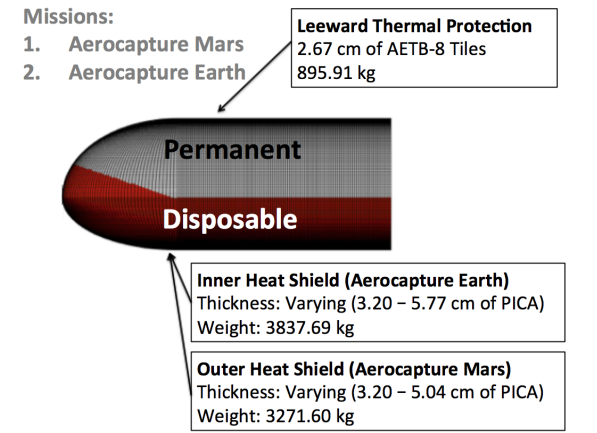


Figure 31: TPS Configuration for the Orbiter Module

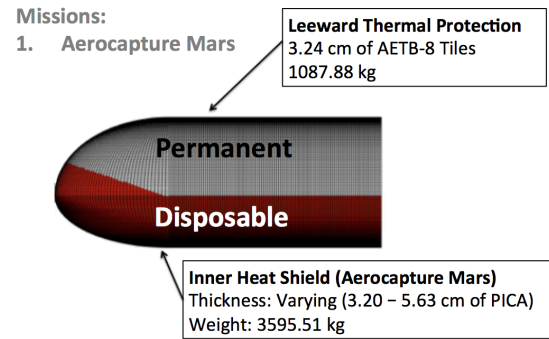


Figure 32: TPS Configuration for the Propellant Module

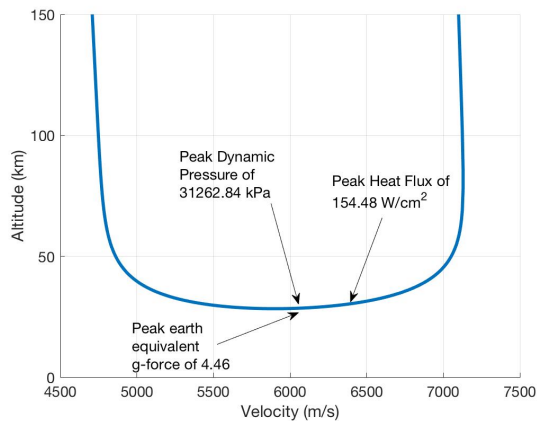


Figure 33: Mars Aerocapture Trajectory for Habitat Module

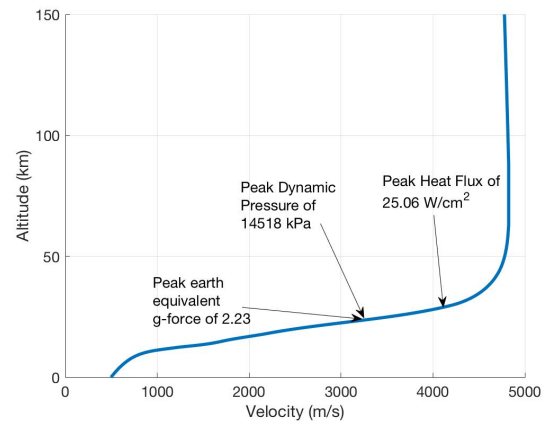


Figure 34: Mars EDL Trajectory for Habitat Module

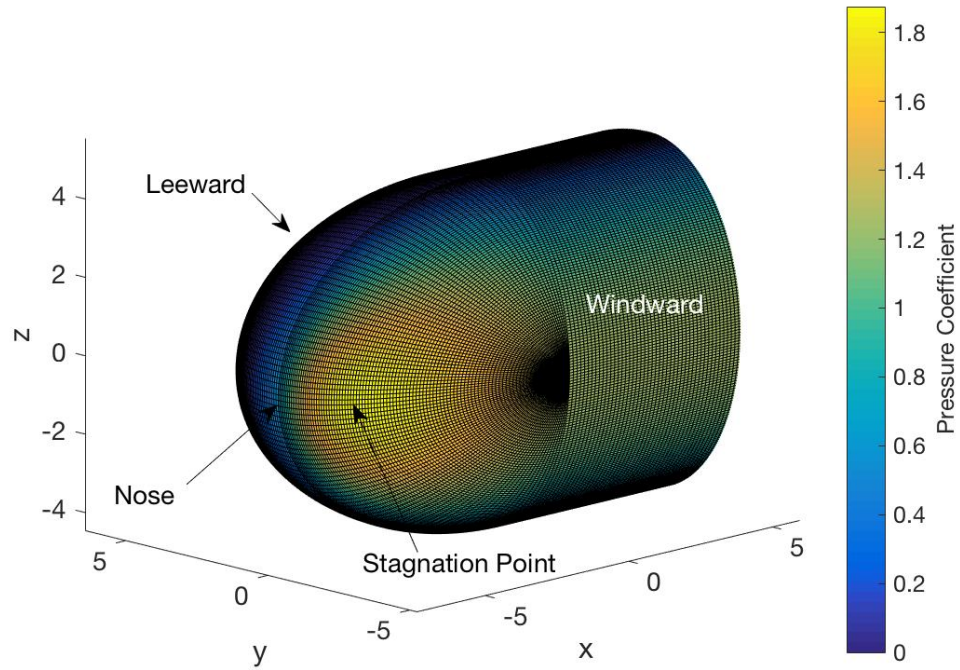


Figure 35:  $C_P$  Distribution for Final Design

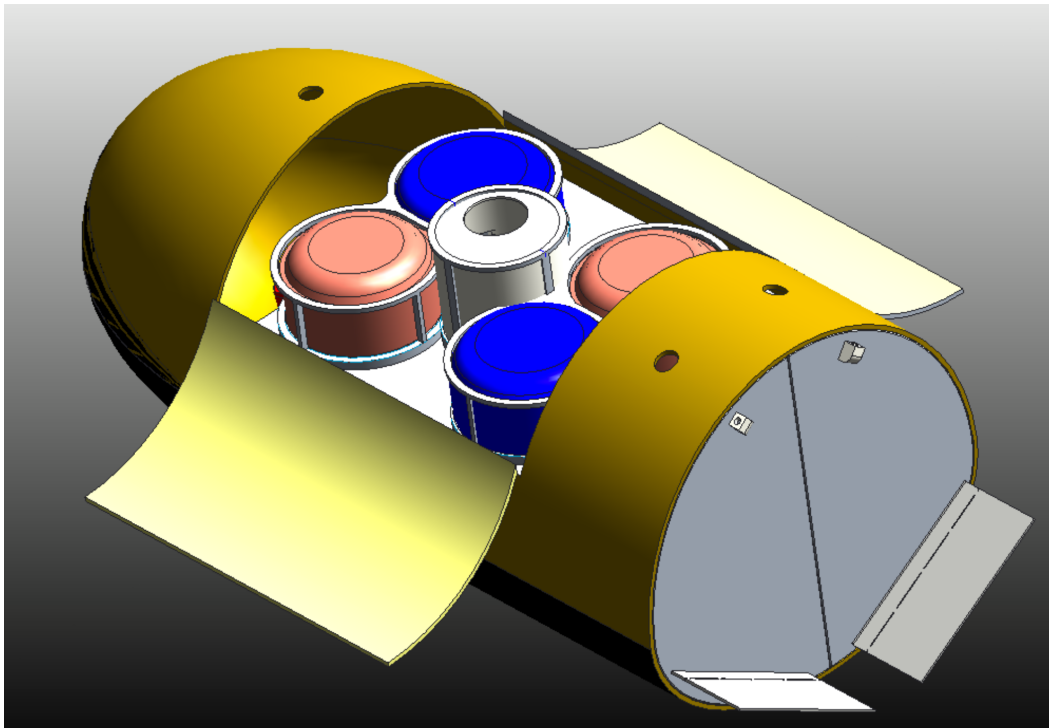


Figure 36: MAV within the Final Design [44]